

Course Code : ECT 3003

UVXV/MW – 24 / 1525

**Third Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DIGITAL SYSTEM DESIGN WITH HDL

Time : 3 Hours]

[Max. Marks : 50

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
-
1. An autonomous system is to be designed that will sound an alarm under certain conditions. The alarm is to sound if the seat belt is not fastened and engine is running, or if the key is in the ignition, the engine is not running, and the driver's door is open. Determine the number of input and output variables, assign names to variables, construct a truth table, and design the system. 10(CO2)

 2. (a) Write Verilog module for all the primitives using conditional operator. 6(CO1)

(b) Write Verilog statements to create :
 - (i) 1k memory locations with 8-bit storage capacity.
 - (ii) An instance of 32 AND gates. 4(CO1)

 3. (a) Design a digital system for 4:1 multiplexer using 2:1 UDP. 6(CO4)

(b) Design a digital circuit using Verilog that detects negative binary number. Consider 8-bit input binary sequence with signed magnitude representation. 4(CO4)

4. Design a sequence detector to detect an overlapping sequence "1001" using Verilog. 10(CO3)
5. (a) Explain architecture of PLA. 6(CO5)
- (b) Demonstrate function implementation in PLA with the help of a digital circuit with a minimum of 3 inputs and 3 outputs. 4(CO5)

